

Overview

The code in this replication package constructs all tables and figures in 'Variety-Based Congestion in Online Markets: Evidence from Mobile Apps,' by Daniel Ershov (AEJ:Micro) using Python and Stata. The Python file "aej_apps_master_python.py" executes 8 Python files, which clean and prepare the data, as described below. The Stata file "aej_apps_master_stata.do" executes 19 Stata files, which perform additional data cleaning and execute the analysis. The Stata files produce 5 of the 6 tables in the main text (Table 1 has no data), and the 18 tables and 10 figures in the online appendix. The replicator should expect the entire code to run for at least one week, and involve a substantial amount of computer memory.

Data Availability and Provenance Statements

Statement about rights:

- I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Summary of Availability:

- Data **cannot be made** publicly available.

Details on each Data Source:

- *App Data:*

App data was purchased from AppMonsta.com in 2015, and is owned by AppMonsta.com. The data is located on the AppMonsta servers and is accessible via an API. This data consists of .json files which include daily snapshots of the US Google Play app store. Each .json file is titled with the date (i.e., the file for January 1, 2014 is titled "20140101.json"). These data need to be downloaded into the directory ".../replication_package/raw_app_data/" using the Python script "appmonsta_1_download.py". This Python script accesses the AppMonsta API (<https://api.appmonsta.com/v1/stores/android/details...>) and runs as part of "aej_apps_master_python.py" (see below).

- *App Ranking Data:*

Data was downloaded from flurry.com in 2015 with a subscription, and is owned by Flurry.¹ This data consists of .csv files which include rankings for the top 500 games and non-games (separately) in each category on the US Google Play app store for a given date. The dates used in this paper are each Friday from Friday January 6, 2012 to Friday, December 26, 2014, and approximate the download-based ranking of apps for that week. Each .csv file is titled with the date (i.e., the file for games on January 1, 2014 is titled "toprankings_games_2014-1-07.csv"). These data need to be downloaded into the directory ".../replication_package/ranking_data_games" and ".../replication_package/ranking_data_nongames".

¹ Identical historical ranking data is available (with subscription) from the following providers: SensorTower, PrioriData, 42Matters, and data.ai.

- *Google Trends Data:*

Data was downloaded from Google Trends (<https://www.google.com/trends>) in 2019. This data consists of 3 .csv files, which include weekly 2012-2014 Google Trends data for the US search terms (1) "Android Games" and "Android Apps", (2) "Google Play Games" and "iOS Games", and (3) "Google Play Games" and "Google Play Apps". .csv file should be stored in the directory ".../replication_package/google_trends". In the code, they are referred to as "AndroidGamesApps_SearchTrends.csv", "GooglePlayGames_iOS_Games_SearchTrends.csv" and "GooglePlayGamesApps_SearchTrends.csv".

Computational Requirements

Software requirements:

1. Python 3.8 or above (with packages numpy, pandas sklearn and parquet installed)
2. Stata 16 or above (with packages ftools, gtools, reghdfe, ivreghdfe, outreg2, estout, sutex2, egenmore and parmest installed)

Memory and runtime requirements:

The data files involved are very large and data cleaning consumes substantial computer memory. A computer with at least 10 cores, >64GB RAM and at least 2 TB memory is recommended. Approximate time needed to reproduce all the analyses from the raw data on a standard 2021 desktop machine is at least one week.

Description of Programs/Code and Instructions to Replicators

The data are cleaned and assembled using Python code below, in order. The processed data are then analyzed using Stata code below. I describe what the replicator needs to do in each of the programs ahead of running the program files. All program files are located in the folder ".../replication_package/".

Data Cleaning/Preparation (in Python):

Python Instructions

In the `aej_apps_master_python.py` file, change the first line to replace "PATH_OF_CURRENT_FOLDER" by the current directory (.../replication_package/). Then execute the script in Python. The file executes the following scripts, in order. Note, the data files involved are very large and data cleaning consumers substantial computer memory. A computer with at least 64GB RAM is recommended.

`python_1_download.py:`

- downloads .json app data from AppMonsta API into the ".../replication_package/raw_app_data/" folder.

python_2_monthly_data_creation.py:

- creates monthly .csv snapshots of all apps in the Google Play store from the .json files, and saves them into the ".../replication_package/monthly_app_data_csv/" folder.

python_3_weekly_data_creation.py:

- creates weekly .csv snapshots of all games and non-games in the Google Play store (separately) from the .json files. Saves each snapshot into the folder ".../replication_package/weekly_app_data_games/" for games, and ".../replication_package/weekly_app_data_nongames/" for non-game apps.

python_4a_file_size.py:

- creates monthly .csv snapshots of app file size for all apps in the Google Play store from the .json files, and saves them into the ".../replication_package/monthly_app_data_csv/" folder.

python_4b_screenshot.py:

- creates monthly .csv snapshots of the number of screenshots and number of video previews for all apps in the Google Play store from the .json files, and saves them into the ".../replication_package/monthly_app_data_csv/" folder.

python_4c_app_ratings.py:

- creates monthly .csv snapshots of the ratings histogram (i.e., N 1 star ratings, N 2 star ratings,..., N 5 star ratings) for all apps in the Google Play store from the .json files, and saves them into the ".../replication_package/monthly_app_data_csv/" folder.

python_4d_related.py:

- creates monthly .csv snapshots of the "related apps" (as defined by Google) for all apps in the Google Play store from the .json files, and saves them into the ".../replication_package/monthly_app_data_csv/" folder.

python_5_reclassification.py:

- As described in Online Appendix B1, this file uses a Random Forest classifier algorithm to re-classify pre-March 2014 game categories into post-March 2014 game categories for each game based on their description. Saves monthly files with the name of the app, their "original category" and their "new category" into the ".../replication_package/monthly_app_data_csv/" folder.

Data Processing and Analysis (in STATA):

STATA Instructions

In the aej_apps_master_stata.do file, change the first line to replace "PATH_TO_CURRENT_FOLDER" with the current directory (.../replication_package/). Then execute the script in STATA. The script then executes the following STATA files, in order. All intermediary and final data files are saved into the folder .../replication_package/processed/. All tables are saved into the

folder ../replication_package/tables/. All figures are saved into the folder ../replication_package/figures/.

stata_1_related_apps_list.do:

- takes the monthly "related apps" files (created using file python_4d_related.py), combines them, and saves them to the folder folder ../replication_package/processed. These files help clean an issue in the data for Sports Games / Sports apps in March and April 2014 (see detailed description in the next .do file).

stata_2_monthly_app_data_creation.do:

- takes the monthly .csv files created using python_2_monthly_data_creation.py, combines and cleans them into one .dta file: monthly_app_data.dta in the folder ../replication_package/processed. As part of the processing, app names are replaced with numerical name_id (the dictionary is saved in the file ../replication_package/processed/name.dta), and developer names are replaced with a numerical dev_id (the dictionary is saved in the file ../replication_package/processed/dev_id.dta). This file also cleans up the bug in the data that does not assign a "Game/Non-Game" status to Sports Games and Sports apps in March and April 2014 (see lines 113-216 in the .do file for a detailed description of this process).

stata_3a_n_photos_videos.do:

- takes the monthly .csv files created using python_4b_screenshot.py, combines and cleans them into one .dta file : n_screenshots_videos.dta in the folder ../replication_package/processed. This file is ready to be merged into monthly_app_data.dta.

stata_3b_app_ratings.do:

- takes the monthly .csv files created using python_4c_app_ratings.py, combines and cleans them into one .dta file : app_ratings.dta in the folder ../replication_package/processed. This file is ready to be merged into monthly_app_data.dta.

stata_3c_app_sizes.do:

- takes the monthly .csv files created using python_4a_file_size.py, combines and cleans them into one .dta file : app_sizes.dta in the folder ../replication_package/processed. This file is ready to be merged into monthly_app_data.dta.

stata_4_reclassifying_game_cats.do:

- takes the monthly .csv files created using python_5_reclassification.py, combines and cleans them into one .dta file : pre_exit_ML_categories.dta in the folder ../replication_package/processed. This file contains the machine learning classified app-types for apps that exited the market before March 2014. It is ready to be merged into monthly_app_data.dta.

stata_5_weekly_data_creation.do:

- takes the weekly .csv files created using python_3_weekly_data_creation.py, combines and cleans them into two .dta file : weekly_nongames_data.dta, for non-game apps, and weekly_games_data.dta for game apps. Both are in the folder ../replication_package/processed.

stata_6a_ranking_data_creation_games.do, and
stata_6b_ranking_data_creation_nongames.do:

- boh of these files do the same thing. Each takes the ranking .csv files from the folder ../replication_package/ ranking_data_games/ or ../replication_package/ ranking_data_nongames/ (depending whether it's 6a or 6b), combines and cleans them into a single .dta file: android_toprankings_games_weekly.dta, for game apps, and android_toprankings_nongames_weekly.dta for game apps. Both are in the folder ../replication_package/processed, and are ready to be merged into the weekly data files created by stata_5_weekly_data_creation.do.

stata_7a_predicting_downloads_games.do:

- merges the android_toprankings_games_weekly.dta file with the weekly_games_data.dta file, and following the description in Online Appendix B.2 creates a file with monthly predicted downloads for all games in the sample: monthly_downloads_games.dta. This file is stored in the folder ../replication_package/processed. This .do file also creates Figure B1, row 1 of Table B2, and the first two columns of Table B3 in the Online Appendix.

stata_7b_predicting_downloads_nongames.do:

- merges the android_toprankings_nongames_weekly.dta file with the weekly_nongames_data.dta file, and following the description in Online Appendix B.2 creates a file with monthly predicted downloads for all non-games in the sample: monthly_downloads_games.dta. This file is stored in the folder ../replication_package/processed. This .do file also creates Table B1, row 2 of Table B2, and the last column of Table B3 in the Online Appendix.

stata_8_preparing_RF_data.do:

- this file prepares monthly data for the reduced form analysis, merging monthly_app_data.dta with files created in the .do files stata_3a, stata_3b, stata_3c, stata_4 and stata_7a and stata_7b, and aggregates the data to the game/non-game or app-type level. The resulting aggregate data file is saved as RF_data.dta in the folder ../replication_package/processed.

stata_9_RF_regressions_main_text.do:

- this file estimates the reduced form regressions described in Section 4.1, and in Section 5.3.1. It produces Table 3 and Table 5 in the main text.

stata_10_model_main_text.do:

- this file performs the model-based empirical analysis described in Section 5 in the main text, producing Tables 4 and 6 in the main text.

stata_11_model_main_text.do:

- this file calculates summary statistics for the various samples used in the paper and creates Table 2 in the main text.

stata_12_RF_regressions_appendix.do:

- this file uses monthly app data to perform the various reduced form robustness checks and alternative specifications discussed in Sections C.1-C.8 of the Online Appendix. It produces Online Appendix Tables B4, C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, and Online Appendix Figures C1, C2, C3, and C4.

stata_13_google_trends.do:

- this file takes the Google Trends .csv files from the folder ../replication_package/google_trends/, and performs the analysis described in Section C.9 of the Online Appendix. It creates Online Appendix Figure C5.

stata_14_developer_switching_analysis.do:

- this file uses monthly app data to perform the analysis described in Section C.10 of the Online Appendix. It creates Online Appendix Figure C6.

stata_15_model_appendix.do:

- this file performs the various model-based robustness checks, and presents additional model-based results using the main text specification. It creates Online Appendix Tables D1, D2 and D3, and Online Appendix Figures D1, D2 and D3.

List of Tables and Code:

The provided code reproduces all data-based tables in the main text of the paper (Tables 2-6, since Table 1 is not data based), and all the figures and tables in the Online Appendix (Tables B1-B4, C1-C11, D1-D3, and Figures B1, C1-C6 and D1-D3).

Table/ Figure #	Program	Line Number
Table 1	N.A. (no data)	
Table 2	stata_11_summary_stats.do	85, 97, 98, 119,120
Table 3	stata_9_RF_regressions_main_text.do	130, 135, 244
Table 4	stata_10_model_main_text.do	500, 512, 524, 538
Table 5	stata_9_RF_regressions_main_text.do	275, 278
Table 6	stata_10_model_main_text.do	820-827
Table B1	stata_7b_predicting_downloads_nongames.do	15
Table B2	stata_7a_predicting_downloads_games.do and stata_7b_predicting_downloads_nongames.do	95 for games 66 for non-games
Table B3	stata_7a_predicting_downloads_games.do and stata_7b_predicting_downloads_nongames.do	100 and 110 for games and

		110 for non-games
Table B4	stata_12_RF_regressions_appendix.do	719-720
Table C1	stata_12_RF_regressions_appendix.do	223-237
Table C2	stata_12_RF_regressions_appendix.do	243-253
Table C3	stata_12_RF_regressions_appendix.do	257-273 And 649 for last column
Table C4	stata_12_RF_regressions_appendix.do	284-323
Table C5	stata_12_RF_regressions_appendix.do	332-388
Table C6	stata_12_RF_regressions_appendix.do	398-477
Table C7	stata_12_RF_regressions_appendix.do	678
Table C8	stata_12_RF_regressions_appendix.do	751-754
Table C9	stata_12_RF_regressions_appendix.do	760-769
Table C10	stata_12_RF_regressions_appendix.do	781-805
Table C11	stata_12_RF_regressions_appendix.do	686-690
Table D1	stata_15_model_appendix.do	518
Table D2	stata_15_model_appendix.do	358
Table D3	stata_15_model_appendix.do	698-728
Figure B1	stata_7a_predicting_downloads_games.do	12
Figure C1	stata_12_RF_regressions_appendix.do	857-868
Figure C2	stata_12_RF_regressions_appendix.do	604-610
Figure C3	stata_12_RF_regressions_appendix.do	944-988
Figure C4	stata_12_RF_regressions_appendix.do	1009-1015
Figure C5	stata_13_google_trends.do	17-65
Figure C6	stata_14_developer_switching_analysis.do	69
Figure D1	stata_15_model_appendix.do	554
Figure D2	stata_15_model_appendix.do	687
Figure D3	stata_15_model_appendix.do	991-999

Acknowledgements:

None.